

① Population vs Sample vs Parameter vs Estimator

- Population $\rightarrow$ entire data
- Parameter (θ) $\rightarrow$ fixed but unknown number describing population, ex: μ, σ^2 , Probability (p).

• Sample $\rightarrow x_1, x_2, \dots, x_n$

• Estimator $\rightarrow$ func. of sample used to estimate parameter

random variable

$$\hat{\theta} = g(x_1, x_2, \dots, x_n)$$

constant

• Sample $n = N$ • Sample var $= \frac{\sigma^2}{n}$ • Sample S.D $= \frac{\sigma}{\sqrt{n}}$

② Error in Estimation $\rightarrow$

$$\text{Error} = \hat{\theta} - \theta$$

$\downarrow$ Estimator
 $\downarrow$ Parameter

Chebyshev bound:

$$P(|\text{Error} - E[\text{error}]| > \delta) \leq \frac{\text{Var}(\text{error})}{\delta^2}$$

Q. Let $x_1, x_2, \dots, x_n$ be i.i.d. Poisson (μ) with unknown parameter μ . An estimator of the mean μ is given by:

$$\hat{\mu} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

use Chebyshev's inequality to find the min. value of n such that $P(|\text{error}| > \frac{\sqrt{\mu}}{10}) \leq 0.01$

sol ① if $x_1, x_2, \dots, x_n$ are i.i.d. Poisson(μ) $\text{Var}(x_i) = \mu$

(A) = var of Sum

$$\text{var}(x_1 + \dots + x_n) = n\mu$$

(B) var of avg: $\text{var}(\hat{\mu}) = \text{var}\left(\frac{1}{n} \sum x_i\right) = \frac{1}{n^2} \cdot n\mu = \frac{\mu}{n}$

$$P\left(\left|\hat{\mu} - \mu\right| > \frac{\sqrt{M}}{10}\right) \leq \frac{M/n}{\left(\frac{\sqrt{M}}{10}\right)^2}$$

$$0.01 = \frac{M/n}{M/100}$$

$$0.01 = \frac{100}{n}$$

$$n = 10,000$$

$$\frac{M/n}{\left(\frac{\sqrt{M}}{10}\right)^2} = 0.01$$

③ # Bias of Estimator

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

↙
expected value of error

- Unbiased estimator: $E[\hat{\theta}] = \theta$
- Biased estimator: $E[\hat{\theta}] \neq \theta$

④ Variance of Estimator $\rightarrow \text{Var}(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$

⑤ Risk (Mean Squared Error - MSE)

$$\text{Risk}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

$$\star \boxed{\text{MSE} = \text{Bias}^2 + \text{Variance}}$$

$(\hat{\theta}, \theta) \quad (\hat{\theta}, \theta) \quad (\hat{\theta})$

$$E[(\hat{\theta} - \theta)^2] = E[\hat{\theta} - \theta]^2 + E[(\hat{\theta} - E[\hat{\theta}])^2]$$

Q. Let $X_1, X_2, \dots, X_n \sim \text{i.i.d. Normal}(\mu, \sigma^2)$. An estimator $\hat{\sigma}^2$ for σ^2 is given by $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2$. Find the bias of $\hat{\sigma}^2$.

Sol.

$$\text{Bias}(\hat{\sigma}^2) = E[\hat{\sigma}^2] - \sigma^2$$

$$E[\hat{\sigma}^2] = E\left[\frac{1}{n} \sum X_i^2 - \bar{X}^2\right] = \frac{1}{n} \sum E[X_i^2] - E[\bar{X}^2]$$

$$\Rightarrow E[X_i^2] \rightarrow \text{var}(X_i) = E[X_i^2] - (E[X_i])^2$$

So,

$$E[X_i^2] = \sigma^2 + \mu^2$$

$$\text{Thus: } \boxed{\frac{1}{n} \sum E[X_i^2] = \sigma^2 + \mu^2}$$

$$\Rightarrow E[\bar{X}^2] \rightarrow \text{Use: } E[\bar{X}^2] = \underbrace{\text{var}(\bar{X})}_{\sigma^2/n} + \underbrace{(E[\bar{X}])^2}_{\mu^2}$$

$$\boxed{E[\bar{X}^2] = \frac{\sigma^2}{n} + \mu^2}$$

$$E[\sigma^2] = (\sigma^2 + \mu^2) - \left(\frac{\sigma^2}{n} + \mu^2 \right)$$

$$\text{Bias} = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\left(\frac{\sigma^2}{n} \right) \Rightarrow \boxed{\sigma^2 \left(1 - \frac{1}{n} \right) = \left(\frac{n-1}{n} \right) \sigma^2}$$

Bias - Variance Trade off

- Low Bias $\rightarrow$ may have high variance
 - Low Variance $\rightarrow$ may introduce bias.
- } Goal: Minimize MSE

Moments of Distribution

① Population Moment $\rightarrow$ expected value of x^k

- $\text{Ber}(p) \rightarrow E[X] = p$
- $\text{Geo}(p) \rightarrow E[X] = \frac{1}{p}$
- $\text{Poisson}(p) \rightarrow E[X] = \lambda$
- $\text{Exp}(p) \rightarrow E[X] = \frac{1}{\lambda}$
- $\text{Normal}(\mu, \sigma^2) \rightarrow E[X] = \mu, E[X^2] = \mu^2 + \sigma^2$
- $\text{Gamma}(\alpha, \beta) \rightarrow E[X] = \alpha/\beta, E[X^2] = \alpha^2/\beta^2 + \alpha/\beta^2$

$E(X) \rightarrow 1^{\text{st}} \text{ moment}$
 $E(X^2) \rightarrow 2^{\text{nd}} \text{ moment}$
 $E(X^3) \rightarrow 3^{\text{rd}} \text{ moment}$

② Sample Moment

$$M_k(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n x_i^k$$

1st Sample moment $\hat{=}$
 $M_1 = \bar{x}$

Methods of moment (MoM)

→ Population moment = Sample moment

↳ steps:

1. Compute theoretical moment $E[X]$
2. set $E[X] = \bar{X}$
3. Solve for parameter

Q. The density func of a cont. rand. var. is given by
 $f_X(x) = \frac{1+\theta x}{2}$, $-1 \leq x \leq 1$

Let $X_1, X_2, X_3, \dots, X_n$ be i.i.d samples of X , find the estimator of θ using methods of moments.

Sol. Pdf: $f_X(x) = \frac{1+\theta x}{2}$, $-1 \leq x \leq 1$

$$E[X] = \int_{-1}^1 x \cdot \left(\frac{1+\theta x}{2}\right) dx = \frac{1}{2} \int_{-1}^1 (x + \theta x^2) dx$$

$$\Rightarrow \frac{1}{2} \times \theta \cdot \frac{2}{3} = \left(\frac{\theta}{3}\right)$$

$$\frac{\theta}{3} = \bar{X}$$

$$\theta = 3\bar{X} \rightarrow \hat{\theta} = \frac{3(X_1 + X_2 + \dots + X_n)}{n}$$

10 # Likelihood Function

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

o Maximum Likelihood (ML) estimation (MLE)

$$\theta_1^*, \theta_2^* \dots \rightarrow \arg \max_{\theta_1, \theta_2} \prod_{i=1}^n f_X(x_i; \theta_1, \theta_2)$$

argument
 (for what value of θ_1 and θ_2 maximum likelihood occur)

↑ Likelihood
 → maximise function over parameter θ_1, θ_2

ex:

Coin tossed 100 times → Land head 62 times

$$\text{Likelihood Func. for } \theta \rightarrow L(\theta) = \theta^{62} (1-\theta)^{100-62} = \theta^{62} (1-\theta)^{38}$$

Q.
 Sol.

Consider $X_1, X_2, \dots, X_n \sim \text{i.i.d. Exp}(\lambda)$. Find the likelihood func. for λ .

Since sample are i.i.d

$$f_X(x) = \lambda e^{-\lambda x}, x \geq 0$$

$$\rightarrow L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i}$$

$$\Rightarrow \text{Simplify: } L(\lambda) = \lambda^n \cdot e^{-\lambda (x_1 + x_2 + \dots + x_n)}$$

★ Q. Suppose that the lifetime of light bulbs is modelled by an exponential distribution with unknown parameter λ . We test 5 bulbs and find that their lifetime of 3, 4, 2, 5, and 6 yrs, respectively. What is the Max. Likelihood estimate for λ ?

Sol.

$$f(x) = \lambda e^{-\lambda x} ; \exp(1) \text{ with pdf in iid}$$

$$(1) L(\lambda) = \lambda^n e^{-\lambda \sum x_i}$$

$$\log L(\lambda) = n \log \lambda - \lambda \sum x_i$$

Differentiate (2) $\frac{d}{d\lambda} \log L(\lambda) = \frac{n}{\lambda} - \sum x_i$

$$\hookrightarrow \text{Set to 0: } \frac{n}{\lambda} = \sum x_i$$

$$(3) \lambda = \frac{n}{\sum x_i} \rightarrow \frac{5}{3+4+2+5+6=20} = \frac{5}{20} = 0.25$$

Q. The density func. of a continuous rand. var. X is given by $\theta k \theta x^{-(\theta+1)}$, $x > k$

where $k > 0$ is a constant. For n i.i.d. samples $X_1, X_2, \dots, X_n$ taken from X , find the maximum likelihood estimate of θ .

Sol.

$$f(x) = \theta k \theta x^{-(\theta+1)}, \quad x > k$$

(Step: 1) Likelihood for iid sample:

$$L(\theta) = \prod_{i=1}^n \theta k \theta x_i^{-(\theta+1)}$$

$$= \theta^n k^n \prod_{i=1}^n x_i^{-(\theta+1)}$$

(Step: 2) Take log

$$\log L(\theta) = n \log \theta + n \log k - (\theta+1) \sum \log x_i$$

(Step: 3) Differentiate

$$\frac{d}{d\theta} \log L(\theta) = \frac{n}{\theta} + n \log k - \sum \log x_i$$

$$\text{Set } = 0 : \frac{n}{\theta} = \sum \log x_i - n \log k$$

$$(Step: 4) \theta = \frac{n}{\sum \log x_i - n \log k} = \frac{n}{\ln(x_1 x_2 \dots x_n) - n \ln(k)}$$